

RFNOID: Protecting RFID Motion Privacy via Metasurface

Yanni Yang*, Zheng Shi*, Zhenlin An[†], Runyu Pan*, Yanling Bu[‡], Guoming Zhang*, Pengfei Hu*, Jiannong Cao[§]

*Shandong University, [†]University of Pittsburgh, [‡]Nanjing University of Aeronautics and Astronautics,

[§]The Hong Kong Polytechnic University

Abstract—RFID tags have become prevailing human motion sensors in applications, like smart homes and health monitoring. However, RFID’s powerful through-wall sensing capabilities raise significant privacy concerns about leaking human motion information, which the RFID sensing community has overlooked. To fill this gap, we propose RFNOID, the first system that can protect human motion privacy against adversarial through-wall RFID sensing. The key enabler is our design of an RFID metasurface consisting of multiple 1-bit phase shifters that can obfuscate motion information. As a pioneering work, we perform theoretical modeling and exploratory studies of the metasurface effect on RFID signals in temporal and spectral domains. Based on the preliminary analysis, we find it is non-trivial to achieve effective signal obfuscation in both domains due to the trade-off in simultaneously increasing temporal signal chaos and masking the human motion spectrum. To tackle this issue, we judiciously devise a metasurface controlling strategy by jointly optimizing the signal entropy, variance, and spectrum distribution to balance the temporal and spectral motion obfuscation. Extensive experiments demonstrate that RFNOID can significantly decrease the adversarial through-wall motion detection rate to less than 6% and increase the respiration estimation error by over 3 $\times$.

Index Terms—RFID sensing, metasurface, wireless security

I. INTRODUCTION

Today RFID-based human sensing systems have surged in popularity, revolutionizing various aspects of our lives, such as intrusion detection [1–3], localization and tracking [4–6], health monitoring [7–9], and activity recognition [10–12]. However, the widespread adoption of low-cost RFID tags and the richness of their sensing capabilities, particularly their through-wall sensing ability, render non-negligible privacy concerns [13–15]. Numerous studies have shown successful attempts in through-wall human motion [3–5] and speech eavesdropping [15–18] via RFID tags for adversarial uses. Besides, the physiological and behavioral features [9, 19] captured by RFID signals for human authentication can be easily intercepted via malicious sniffing. Therefore, it is imperative to protect RFID-based human motion sensing systems from unauthorized access and malicious attacks.

However, existing literature lacks an efficient way to safeguard RFID motion sensing. Current defenses to communication eavesdropping, such as channel randomization [20], require customized RFID readers, making them incompatible with existing systems. Furthermore, recent solutions for protecting other wireless sensing systems, including FMCW radar [21] and WiFi [22, 23], cannot be directly applied to RFID due to essential differences in protocols and sensing models. In short, we desire a new method to defend against privacy intrusions in RFID-based motion sensing systems.

*Runyu Pan is the corresponding author.

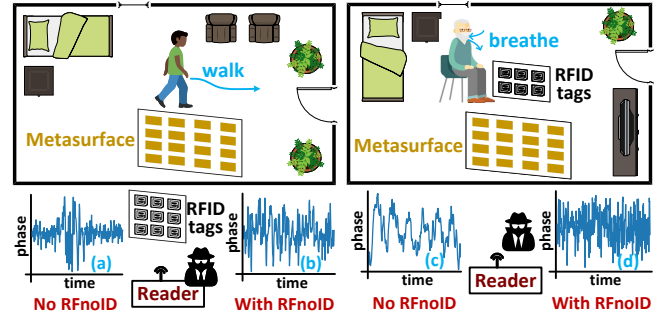


Fig. 1. Protecting through-wall RFID human motion privacy with RFNOID. Our metasurface can successfully obfuscate walking and breathing patterns.

Thus, we present RFNOID, the first system to protect human motion privacy from adversarial through-wall RFID sensing. The key idea is to design a programmable metasurface for obfuscating the adversary’s RFID channel. Different from active signal jamming methods [20], the metasurface only *passively* reflects RFID signals, thereby reducing system complexity. We also employ a *low-cost* metasurface design with the simple yet effective 1-bit phase shifter (i.e., 0 or π) unit, in which a PIN-diode is switched on or off to control the signal phase. The devised metasurface is configured to be *compatible* with current RFID systems for seamless integration. By deploying and controlling the metasurface in the sensing area, part of the RFID signals will be reflected by the metasurface units with different on/off states and superimpose with the signal reflected by the human body. In other words, the metasurface acts as an irregularly moving reflector to disrupt the adversarial channel. Thus, with signal variations caused by the metasurface stronger than those from the human body, we can hide the human motion pattern from leakage. As depicted in Fig. 1, by applying RFNOID, no matter whether the adversary snoops on the RFID tags outside or inside the target room, the original walking and respiration motion signals are obfuscated. Note that the metasurface facing the wall does not affect the legitimate RFID sensing inside the room.

To achieve the desired spoofing against the adversary, significant challenges in *signaling mechanism* and *obfuscation strategy* must be overcome. Regarding *signaling mechanism*, as a pioneer work, the unclear mechanism of how the metasurface reflected signal affects the through-wall RFID human motion sensing signals must be unraveled to achieve effective signal obfuscation. This includes (1) the metasurface effect on the RFID sensing model and received signal indicator (i.e., amplitude and phase) and (2) the metasurface’s superimposition outcome with the human reflected signal.

In terms of *obfuscation strategy*, the adversary can snoop on human motions from different views, including *temporal*

and *spectral* domains. Thus, the obfuscated RFID signal by the metasurface should overwhelm the human reflected signal in both domains. However, existing obfuscation strategies via metasurface [22, 23] only focus on the temporal signal amplitude but ignore spectral features; besides their strategies that manipulate amplitude cannot be naively transferred to RFID sensing which mainly employs the signal phase. To this end, simultaneously signal phase obfuscation in temporal and spectral domains entails intricate control of the metasurface, which yet lacks a systematic approach.

To tackle the *signaling mechanism* challenge, we first model the effect of the metasurface on the RFID channel and signal propagation indicators to understand the metasurface influence on through-wall RFID motion sensing. Our modeling outcomes reveal that different combinations of the metasurface units' on/off states can incur distinctive phase shifts over the metasurface reflected signal, which superimposes with the human motion signal into an obfuscated version. Then, we systematically analyze the effect of unit states on the RFID signal from both temporal and spectral perspectives, which provides us with key observations for successful RFID signal obfuscation during human motions.

To address the *obfuscation strategy* challenge, we draw on the qualitative and quantitative analysis of the metasurface effect and summarize the key goals of the metasurface controlling strategy for achieving strong signal phase obfuscation in both temporal and spectral domains. That is: the metasurface-affected signal phase should simultaneously have (1) a high signal entropy, (2) a high temporal variance, and (3) dominant frequencies around the human motion frequency band. However, a trade-off exists among these three targets. Thus, we strive to devise a compound metasurface controlling strategy to strike a balance among the above three objectives via theoretical analysis and real-world observations.

In sum, our contributions are listed as follows:

- In this paper, we propose the RFNOID, the first-of-its-kind system for protecting human motion privacy from adversarial through-wall RFID sensing via an efficient metasurface.
- We theoretically model the metasurface effect on the human-reflected RFID signal in temporal and spectral domains. We design a novel metasurface controlling strategy to balance effective temporal and spectral domain obfuscation.
- Extensive experiments show that RFNOID can significantly hamper adversarial RFID motion sensing using representative sensing methods. Typically, the human walking motion detection rate is lower to 6% or less, and the respiration rate estimate error is enlarged by $3\times$.

II. PRIMER ON THROUGH-WALL RFID MOTION SENSING

The RFID signals at the 860 MHz–960 MHz UHF band have a strong ability to penetrate through walls due to their relatively low attenuation. Thereby, extensive studies have exploited RFID signals for through-wall human motion sensing, and they can be roughly divided into two categories. One is through-wall human motion detection without tags in the room [3–5], which follows the signal model as shown in

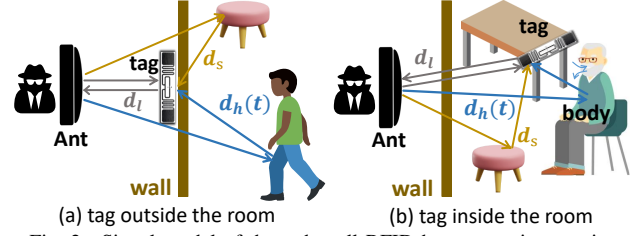


Fig. 2. Signal model of through-wall RFID human motion sensing.

Fig. 2(a). The adversary deploys the RFID antenna (Ant) and tags outside the target room. When the person walks around, the received signal $R(t)$ from the tag can be expressed as:

$$R(t) = \alpha_l e^{-j \frac{2d_l}{\lambda}} + \alpha_s e^{-j \frac{d_s + d_l}{\lambda}} + \alpha_h(t) e^{-j \frac{d_h(t) + d_l}{\lambda}}, \quad (1)$$

where α_l , α_s , and $\alpha_h(t)$ denote the attenuation factor of the line-of-sight (LoS) path signal (in grey) and multipath signals from the surrounding stationary object (in yellow) and the human body (in blue). Accordingly, d_l , d_s , and $d_h(t)$ refer to the length of each signal propagation path. λ is the RFID signal wavelength. Since the RFID tags and stationary objects are static, only the moving human body with a varying $d_h(t)$ changes $R(t)$. Note that the RFID signal amplitude is significantly weakened by the wall. Thus, researchers mainly utilize the phase changes incurred by the body motion: $\Delta\phi_h(t) = \frac{d_h(t)}{\lambda} \bmod 2\pi$ to infer the through-wall motion pattern. As shown in Fig. 1(a), when a person walks around inside the room, it can cause salient phase changes of the tag deployed by the adversary behind the wall, indicating the person's presence inside the room.

The other common through-wall RFID sensing scenario is that RFID tags are already deployed inside the room for certain uses [8, 10, 24]. Under this setup, the adversary is still able to capture the multipath signal reflected by the person (i.e., the signal path Ant-body-tag-Ant), as shown in Fig. 2(b). In Fig. 1(c), we depicted the adversary received phase of the tag placed indoors, which exhibits a periodic pattern of the person's respiration inside the room.

III. THREAT MODEL

RFNOID aims at protecting human motion privacy leaked by adversarial through-wall RFID sensing. Details of the threat model are described as follows.

Adversary setup: The adversary aims to covertly monitor the victim's motions within an enclosed room, such as an office or home. Nonetheless, the adversary fails to access the area neither physically nor visibly, but manages to place the easily obtainable commodity RFID reader and antenna near the target room instead, deploying them in a nearby room close to high-risk areas such as wooden doors or glass walls, where RFID signals are more likely to penetrate, while avoiding locations shielded by dense materials, like thick concrete walls, where signals would be blocked. To obtain the human motion-affected RFID signal, the adversary may (1) place several RFID tags behind the wall, as shown in Fig. 2(a), if no tag is inside the target room, or (2) directly receive the signal from the tag positioned in the room without deploying extra tags, as depicted in Fig. 2(b). In this work, we exemplify two

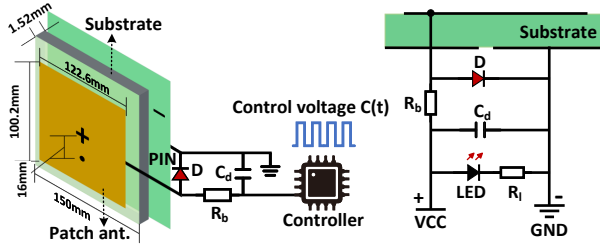


Fig. 3. Fabrication of RFID metasurface unit.

typical adversarial sensing tasks: large-scale walking motion detection and small-scale respiration rate monitoring.

Adversary sensing algorithms: The adversary owns the domain knowledge of the widely used RFID signal processing methods (e.g., phase unwrapping and smoothing methods) and sensing algorithms. For walking motion detection, two typical algorithms are used: (1) the variance-based approach [2, 3]: It is reasonable to assume the target room is static most of the time. Thus, the adversary can pre-collect a long-term signal phase, calculate its variance, and determine a threshold above which denotes motion is detected. (2) the learning-based approach [9, 19, 25]: Hand-crafted features (e.g., statistical features) or neural network latent vector (e.g., auto-encoder) are extracted from the pre-collected signal phase in temporal and spectral domains and fed into a one-class classifier (e.g., support vector machine) for motion detection. For respiration monitoring, the adversary performs the Fast Fourier Transformation (FFT) and bandpass filtering within the human respiration band on the signal phase and looks for the strongest frequency component as the respiration rate [8, 26].

IV. RFID METASURFACE DESIGN AND MODELING

A. Fabrication of the Reflective Metasurface

In RFNOID, we fabricate a low-cost metasurface with N uniformly placed units, as depicted in Fig. 14(a). The detailed composition of each unit is shown in Fig. 3. On the top of the unit is a patch antenna that reflects the UHF RFID signal. In the middle is a substrate followed by the ground plane at the bottom. The key component in the unit is the PIN diode D whose on/off states change the reflection coefficient. Accordingly, when the controller output $C(t)$ switches between low and high voltages, the unit will demonstrate a 180° phase shift, i.e. 1-bit phase shifter between 0 and π , over the incident RFID signal. In our implementation, we optimize the patch antenna size and substrate thickness in the CST studio to ensure an effective 180° phase shift over the UHF frequency band. We also add a biasing resistor R_b and a decoupling capacitor C_d to stabilize the circuit. The LED and a current limit resistor R_l are used to monitor the controller output voltage. We employ Arduino Mega 2560 with 54 output pins as the controller, which can support up to 7×7 sized metasurface. The total metasurface fabrication cost for a 4×4 metasurface is around \$90 USD, which is comparable with the expense of one RFID antenna. Once fabricated, the metasurface is placed in high-risk indoor areas, i.e., behind wooden doors or glass walls.

B. Modeling Metasurface Effect on Human Motion Sensing

To fulfill the objective of obfuscating human motion signals using the devised metasurface, we first model the metasurface

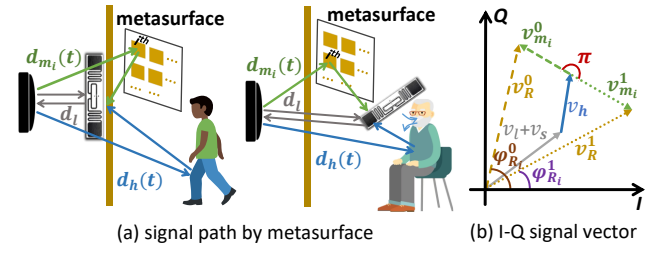


Fig. 4. Effect of metasurface on the received signal phase.

effect to answer two questions: 1) How does the metasurface affect the original RFID sensing signal model? 2) Can the metasurface with only 1-bit phase shifters obfuscate the motion signal phase in various ways?

As depicted in Fig. 4(a), after deploying the metasurface inside the target room, the i^{th} reflective unit, i.e., U_i , introduces a new signal propagation path between the RFID antenna and tag with the attenuation factor of $\alpha_{m_i}(t)$ and length of $d_{m_i}(t)$. Then, the received tag signal becomes:

$$R(t) = \alpha_l e^{-j \frac{2d_l}{\lambda}} + \alpha_s e^{-j \frac{d_s + d_l}{\lambda}} + \alpha_h(t) e^{-j \frac{d_h(t) + d_l}{\lambda}} + \sum_{i=1}^N \alpha_{m_i}(t) e^{-j \frac{d_{m_i}(t) + d_l}{\lambda}}. \quad (2)$$

In Eq. (2), although the path length $d_{m_i}(t)$ does not change physically, it is virtually controlled by the applied voltage to the unit. To understand this effect, we look into the signal vectors in the in-phase and quadrature (I-Q) domain. The received signal vector $v_R^{0/1}$ at low or high voltage is a combination of the LoS tag signal (v_l), human-reflected signal (v_h), and metasurface-reflected signal ($v_{m_i}^{0/1}$) as below:

$$v_R^{0/1} = v_l + v_s + v_h + \sum_{i=1}^N v_{m_i}^{0/1} \quad (3)$$

To interpret Eq. (3), we first look at one unit, e.g., the U_i 's signal vector. Due to the π phase difference between $v_{m_i}^0$ and $v_{m_i}^1$, the received signal phase will change between two results: $\phi_{R_i}^0$ or $\phi_{R_i}^1$. Then, if all N units function together, there can be 2^N different phase results between 0 and 2π . For instance, when $N = 16$, the signal phase can theoretically exhibit $2^{16} = 65536$ unique values within $[0, 2\pi]$. However, due to the superposition of static signal vector v_l , v_s , and dynamic vector v_h from human motions, the actual phase would vary in a smaller range than $[0, 2\pi]$. Thus, the combination of multiple 1-bit phase shifters can incur abundant phase changes to obfuscate those caused by human motions.

C. Preliminary Studies and Observations

Apart from the above theoretical modeling, we conduct experiments to measure real-world metasurface impact on the human motion signal. To succeed in through-wall motion sensing, the adversary would make full use of the received RFID signal from both temporal and spectral domains. Thus, we conduct preliminary studies to investigate the metasurface effect from both temporal and spectral perspectives.

1) *Temporal perspective:* The temporal phase change incurred by the metasurface is desired to have high variances considering the commonly used variance-based approach to detect human motions. From Eq. (3), we conjecture that flipping more units on/off could introduce a larger phase variation.

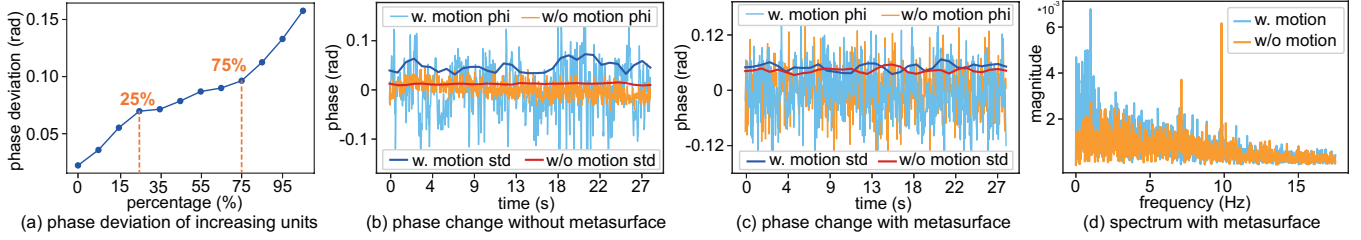


Fig. 5. Temporal perspective: (a) phase deviation for flipping different numbers of units, (b) no metasurface: phase and standard deviation with/without human motions, (c) control metasurface between min- and max-set: phase and standard deviation with/without human motions, (d) spectrum of phase in (c).

To validate this, we increase the percentage of units switched on and off for various sizes of metasurfaces ($N = 9, 16, 25$) to examine the effect on the phase variation. For a certain unit number, we randomly traverse different combinations of units. The unit flipping rate is set at 20 Hz to cover human motion frequencies and below the common RFID signal sampling rate at 30 Hz-50 Hz. The measured average phase deviation of the received tag signal for different percentages of units is shown in Fig. 5(a). It shows that the phase deviation gradually increases with more flipped units, verifying our hypothesis and embodying our first observation as follows.

Observation 1: *The phase variation is positively correlated to the number of units flipped.*

As such, we may control the metasurface by intuitively switching on/off the maximum number of units to attain a high variance. However, such a strategy would result in a periodic temporal pattern, which can be easily filtered in the spectral domain. We tackle this problem by constructing a min-set (0-15% units) and a max-set (85%-100% units), and reciprocally switching on the units between two sets. The units in each set are arbitrarily selected from all units to introduce certain phase randomness. Then, we test this strategy with and without human walking motions and show the signal phase and its deviation in Fig. 5(c). For comparison, the phase change without metasurface is depicted in Fig. 5(b). Without metasurface, the average phase deviation is only 0.006 rad in static environments. While this deviation can be lifted to 0.023 rad (4 $\times$) using the metasurface. Most importantly, the running metasurface makes it difficult to distinguish the phase change and its deviation under human motions from those in static environments, as shown in Fig. 5(c). Thus, this strategy can obfuscate the motion signal in the temporal domain.

However, when the phase in Fig. 5(c) is transformed into the spectral domain, as depicted in Fig. 5(d), we can easily differentiate the spectrum with and without motions from the low-frequency band below 3 Hz, which indeed is the human torso and limb movement frequency band during walking [27]. This is because the way we control metasurface still bears a regular pattern, e.g., a single strong component at 10 Hz as a result of the 20 Hz flipping rate, with *less entropy* at other frequency bands. Besides, the human respiration rate also lies in a similar low-frequency band within 1 Hz. Note that simply decreasing the unit flipping rate for stronger low-frequency components is inadequate because this fails to mask the high-frequency harmonics of human motions. Therefore, using the current metasurface configuration, we face the following issue.

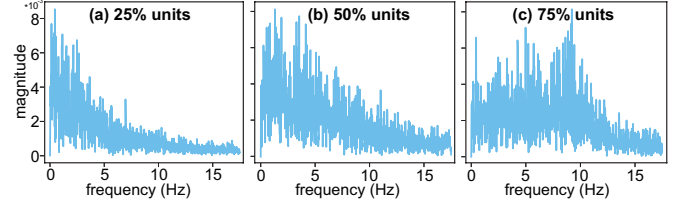


Fig. 6. Phase spectrum of randomly flipping different numbers of units.

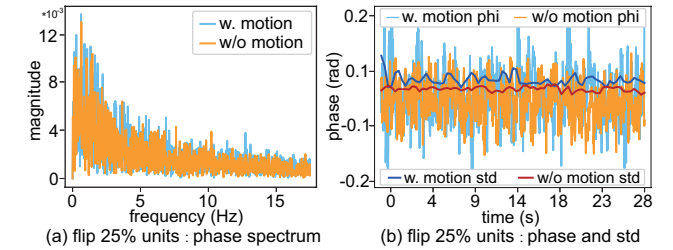


Fig. 7. Randomly flipping 25% units with and without human motions: (a) phase spectrum, (b) raw phase and standard deviation.

Issue 1: *High phase variance alone fails to obfuscate the motion signal in the spectral domain due to low phase entropy in the human motion frequency band.*

Therefore, we also need to consider the human motion pattern in the spectral domain when controlling the metasurface.

2) *Spectral perspective:* Since human motions primarily occur in low frequencies, the metasurface is expected to strengthen the low-frequency components of the affected signal phase. To this end, we analyze the phase spectrum when different numbers of units are flipped. With a 20 Hz unit flipping rate, the maximum frequency of phase variation, i.e., 10 Hz, is achieved by repetitively switching on/off all units. As such, we postulate that if fewer units are randomly selected and flipped each time, lower frequencies (< 10 Hz) could be obtained. To verify this hypothesis, we randomly select and repetitively switch on/off 25%, 50%, and 75% units, respectively, and depict their phase spectrum in Fig. 6. It shows that high-frequency components increase with more units flipped, validating our hypothesis as the second observation.

Observation 2: *The phase component frequency is positively correlated to the number of units flipped.*

We interpret the observation as follows: When flipping fewer units, the gently varying phase demonstrates a slow general change, thus producing more low-frequency components.

Next, we leverage this observation to enhance spectral obfuscation. Fig. 6(a) shows that stronger frequency components below 3 Hz can be attained by using 25% units. Thus, we control the metasurface by randomly flipping 25% units every time and illustrate the corresponding phase spectrum with and

without human walking motions in Fig. 7(a). Now, the two spectra are mixed where the metasurface signal masks the human motion. However, since only 25% units are flipped each time, this strategy leads to insufficient phase variations in the temporal domain. As shown in Fig. 7(b), the phase variations without motions are smaller than those under walking. Thus, the adversary may still detect human motions from the temporal phase, leading to another issue as below.

Issue 2: *Centering the frequency components around human motion frequencies by flipping fewer units alone leads to insufficient signal obfuscation in the temporal domain.*

Respecting **Observation 1 & 2**, we conclude that the units flipped play an important role in obfuscating the RFID signal phase. However, as **Issue 1 & 2** demonstrates, a trade-off exists between temporal and spectral domain obfuscation: temporal domain obfuscation requires a large number of flipped units, while spectral domain obfuscation requires just the opposite. Therefore, a novel approach is required to control the metasurface and strike a proper balance between them.

V. METASURFACE CONTROLLING STRATEGY FOR MOTION SIGNAL OBFUSCATION

Based on the preliminary analysis, a well-designed metasurface controlling strategy is required to successfully obfuscate human motion patterns from both temporal and spectral perspectives and defend against adversary sensing algorithms. To achieve this, we first summarize three goals for the desired metasurface controlling strategy as below:

Goal 1: *Attain a high phase entropy.* The adversary sensing algorithms are designed to extract assured and relevant motion information from the RFID signal phase. To obfuscate motion signals, the metasurface incurred phase ought to attain a high entropy with greater unpredictability and complexity.

Goal 2: *Attain a high temporal phase variance.* The large-scale human motions can cause significant temporal phase changes. The metasurface should be controlled to attain drastic phase variation to overwhelm the motion phase changes.

Goal 3: *Attain strong spectral components at human motion frequencies.* Human motions are dominant in low frequencies. Thus, the metasurface phase spectrum shall attain stronger low-frequency components to mask the motion spectrum.

We then devise and implement a new metasurface controlling strategy to fulfill these goals.

A. Goal 1: Achieving High Phase Entropy

In information theory, entropy is a measure of the signal uncertainty. To confuse the adversary, we should maximize the entropy of the RFID signal phase reflected by the metasurface. As Eq. (3) indicates, the on/off state of each metasurface unit determines the phase change. Therefore, the goal of maximizing the phase entropy can be transformed into maximizing the entropy in selecting the unit state. As such, given a N -sized metasurface, the selection of the unit state entails two subsequent steps: (1) How many units to flip? and (2) Which units to flip? Since the two steps are orthogonal, the maximum entropy is attained by maximizing the entropy of each step.

The above two questions in steps (1) and (2) can be solved by formulating them into a mathematical problem. Specifically,

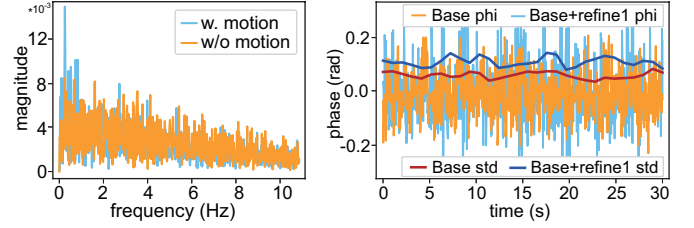


Fig. 8. Phase spectrum when using the Base strategy

step (1) is to select an integer X_n in $[0, N]$, and step (2) is to select a set of units X_s from $C_N^{X_n}$ unit combinations. They are actually the same problem: Given a finite discrete range, what is the maximum entropy distribution? Based on the Principle of Maximum Entropy [28], the answer is *uniform distribution*.

Thus, the number of units X_n to flip in step (1) and the set of units X_s , where $X_s = \{U_i, \dots\}$ and $|X_s| = X_n$, selected from the N -sized metasurface in step (2) should all follow the uniform distribution. To this end, X_n and X_s should be randomly picked with equal probability. Accordingly, we shape a base controlling strategy as follows:

Base strategy: When controlling the metasurface, we first randomly select the number of units X_n to flip from $\{0, 1, 2, \dots, N\}$, then flip a random combination of X_n units among all units.

To generate random numbers with real-world entropy, we utilize the Arduino free-running timer t_r as the seed to generate pseudorandom numbers for X_n and X_s . Then, we apply this strategy to control metasurface units and compare the affected phase spectrum with and without human motions. The orange spectrum in Fig. 8 shows that with maximized entropy, the spectrum without human motions is dispersed among a relatively wide frequency band with no outstanding frequency component. In contrast to the orange spectrum with a strong 10 Hz component in Fig. 5(d), the flattened spectrum can inhibit the adversary from eliminating the metasurface effect using certain filters. However, enhancing the entropy alone with randomly flipped units results in mediocre obfuscation performance in both temporal and spectral domains. The phase variance requires further enlargement, and the phase spectrum demands stronger components around human motion frequencies. Thus, two additional refinements of the Base strategy are introduced in the following sections.

B. Goal 2: Achieving High Phase Variance

As indicated by Observation 1 and Fig. 5(a), larger phase variations are obtained when more units flip their states. Thus, we can attach a higher probability to the simultaneous flipping of a greater number of units in the Base strategy. Specifically, we add one condition that inspects the number of switched-on states n_{on} among all units at the current timestamp. If the current n_{on} is small (e.g., 0, 1, and 2) and we switch on all units at the next timestamp, the number of flipped units becomes larger. Similarly, if the current n_{on} is large (e.g., $N-1$ and N), we can switch off all units to make more units change their states. In this way, the possibility of flipping a large number of units is increased compared with the Base strategy. Meanwhile, random unit flipping is still maintained.

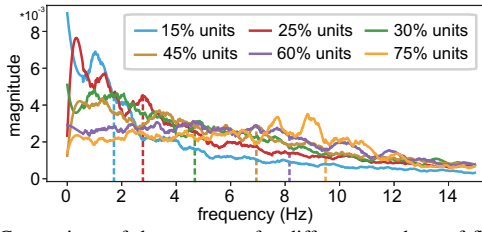


Fig. 10. Comparison of the spectrum for different numbers of flipped units.

To avoid cyclic switch on/off of all units which destroys randomness, we further set a *flag* to cut continuous switching. We summarize the above metasurface configurations to refine the Base strategy as below.

Strategy refinement 1: If a random flip resulted in a n_{on} less than $\sigma_s \times N$ or more than $\sigma_l \times N$ ($\sigma_s < \sigma_l$), we switch on or off all units at the next timestamp respectively. Otherwise, we flip units randomly according to the Base strategy.

In the mechanism above, two thresholds σ_s and σ_l are yet to be assigned, where σ_s is used to find the timestamp when n_{on} is smaller than $\sigma_s \times N$, and σ_l is for detecting the timestamp when n_{on} is larger than $\sigma_l \times N$. To determine σ_s and σ_l , we draw on the phase deviation curve along different percentages of flipped units in Fig. 5(a), where two turning points are detected, i.e., 25% and 75% of all units on the metasurface. The phase deviations below 25% and above 75% units endure prominent decline and incline, respectively. Therefore, by setting σ_s and σ_l to 25% and 75%, we can acquire more significant phase variation in the temporal domain.

We combine the Base strategy and refinement 1 to show the metasurface affected phase and its standard deviation in Fig. 9. Compared with the phase change of the Base strategy, integrating refinement 1 introduces larger phase variances, which are beneficial for obfuscating temporal motion signals.

C. Goal 3: Achieving Strong Spectral Components around the Human Motion Frequency Band

Recall that the entropy enhancement in the Base strategy falls short of obfuscating the spectral human motion pattern in low frequencies. Thus, the third goal is to refine the metasurface controlling strategy so that frequency components can be dominant around the human motion frequency band.

To fulfill this goal, the first step is to elucidate which frequency band the human-reflected RFID signal phase is located. For the large-scale walking motion, the moving torso and limbs have different effects on the RFID signal phase. Suppose the torso moves with a speed of v m/s. Then, it will introduce $\sim 2v$ change of the path length d_h per second. Once the d_h changes over one wavelength of the RFID signal, the phase will fluctuate over 2π ; thus the phase changes with a frequency of $2v/\lambda$. Since the household walking speed is mostly within 0.4 m/s [29], the corresponding frequency band is below 2.5 Hz. At the same time, the limb movement swinging at 1 Hz-2 Hz directly reflects the phase frequency. In terms of small-scale breathing activity, the human respiration rate is mostly around 0.2 Hz-0.5 Hz [30]. In a word, the interested human motions mainly move within 0.2 Hz-2.5 Hz.

Next, we employ Observation 2, which states that flipping fewer units can produce stronger low-frequency components.

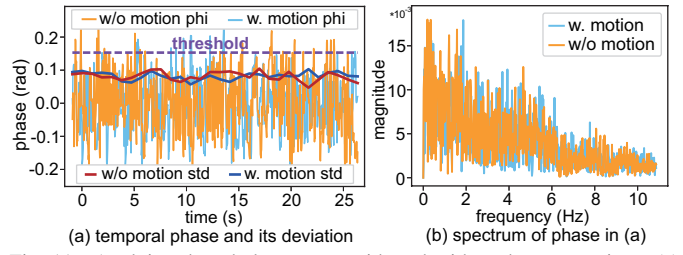


Fig. 11. Applying the whole strategy with and without human motions: (a) temporal phase and its standard deviation, (b) spectrum of phase in (a).

Algorithm 1 Metasurface Controlling Strategy in RFNOID

Require: $N, t_r, \sigma_s, \sigma_l, w, \sigma_f$, all units off
 $srand(t_r)$; $flipSet = []$; $keepSet = [U_1, \dots, U_N]$; $flag = RAND$
while True **do**
 if $n_{on} < \sigma_s \times N$ and $flag = FLIP$ **then** ▷ Refine 2
 $writeToPins(allpins, ON)$; $flag = RAND$;
 else if $n_{on} > \sigma_l \times N$ and $flag = FLIP$ **then**;
 $writeToPins(allpins, OFF)$; $flag = RAND$;
 else
 $randomNum = getRandom(0, 1)$;
 if $randomNum < w$ **then** ▷ Refine 1
 $percentage = getRandom(0, \sigma_f)$; ▷ Base
 else
 $percentage = getRandom(\sigma_f, 1)$;
 end if
 $switchNum = percentage \times N$
 for $i = 0, i < switchNum, i++$ **do** ▷ Base
 $ind = getRandom(0, 1) \% (length(keepSet))$;
 $flipSet.append(U_{ind})$; $keepSet.delete(U_{ind})$;
 end for
 $writeToPins(flipSet, FLIP)$; $flag = FLIP$;
 end if
end while

Thereby, we can boost the chances of flipping a smaller number of units with the following strategy refinement:

Strategy refinement 2: We attach a higher likelihood w to choosing a number less than $\sigma_f \times N$ as the number of units to flip X_n in the Base strategy.

To determine σ_f , we carefully compare the spectrum distribution for different percentages of randomly flipped units on various sizes of metasurface from plentiful field data in Fig. 10. We locate the boundary of the dominant frequency components (above average) highlighted by dashed lines. The boundary gradually grows to larger frequencies with an increasing number of units, again confirming our hypothesis in Observation 2. Since the closest boundary to the human motion frequency band is the red dashed one near 2.5 Hz, we set σ_f to 25%. To decide the weight w , we calculate the spectrum similarity with and without human motions using different w within (0.25, 1), and w is set to 0.6 with the highest similarity.

Metasurface Controlling Strategy as a Whole: We combine the Base strategy with its two refinements into a compound final strategy, as shown in Algorithm 1. The temporal phase and spectrum with and without human walking motions using this strategy are depicted in Fig. 11. It can be observed that: (1) The temporal phase without motions exhibits as significant variations as those under motions in Fig. 11(a); (2) The metasurface-affected phase without motions also successfully overwhelms the phase spectrum with motions in Fig. 11(b).

We further show the effectiveness of our devised strategy

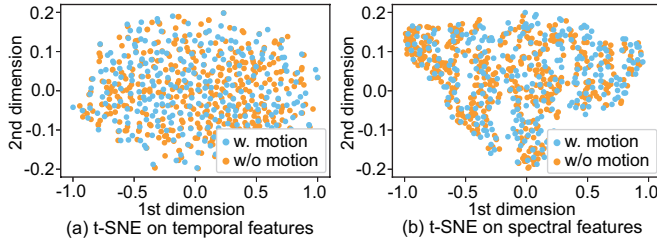


Fig. 12. t-SNE results on statistical features with and without human motions: (a) temporal features, (b) spectral features.

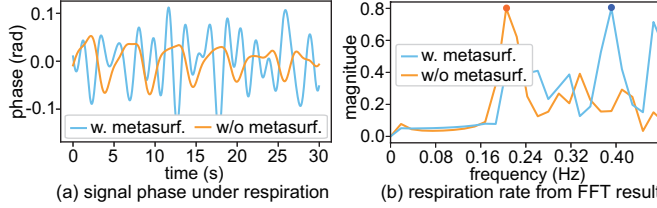


Fig. 13. Respiration monitoring with and without the running metasurface: (a) temporal phase, (b) FFT result of the phase in (a).

using the through-wall motion sensing algorithms mentioned in Section III, including variance-based and learning-based methods. In terms of the large-scale walking motion, we pre-collect a 30 min signal to extract a variance threshold for human motion detection using the method in [3], i.e., the dashed purple line in Fig. 11(a). With RFNOID, the phase variances under human motions are still below the threshold, which makes it difficult for the adversary to detect the inner human motion. We also extract statistical features from the temporal phase and its spectrum and show the feature distribution after dimension reduction via t-SNE in Fig. 12. The features with and without human motions are fully intertwined, which makes it hard to detect the motion by learning-based methods as well. In terms of the small-scale respiration activity, we compare the signal phase after a band-pass filter at 0.2 Hz-0.5 Hz band and the FFT result with and without the running metasurface in Fig. 13. The metasurface obfuscates the original periodic respiration signal with more irregular fluctuations and a large error in respiration rate estimation of 0.15 Hz.

VI. EVALUATION

A. Evaluation Setup

1) *Hardware*: The fabricated RFID metasurface is shown in Fig. 14(a). The model of the substrate and PIN diode are ZYF255DA and SMP1302-079LF, respectively. To avoid oxidation, we lube all units' patch antennas. The Impinj R420 RFID reader and a 12 dBi antenna are utilized as the adversary's attacking devices. To balance the trade-off between the number of used tags and the sampling rate of each tag, we employ six tags into a 3×2 matrix for acquiring sufficient signals to sense human motions. The sampling rate of each tag is around 30 Hz. Different tag models, including ALN-9762 and Impinj H47, are selected in our experiments.

2) *Evaluation environments*: We evaluate the RFNOID performance in three different environments, including (1) one office with a gypsum wall, (2) one meeting room with a wooden wall, and (3) one lab with a glass wall, as shown in Fig. 14(b-c). The deployment of the adversary's RFID antenna, tag matrix, and metasurface for walking motion detection and

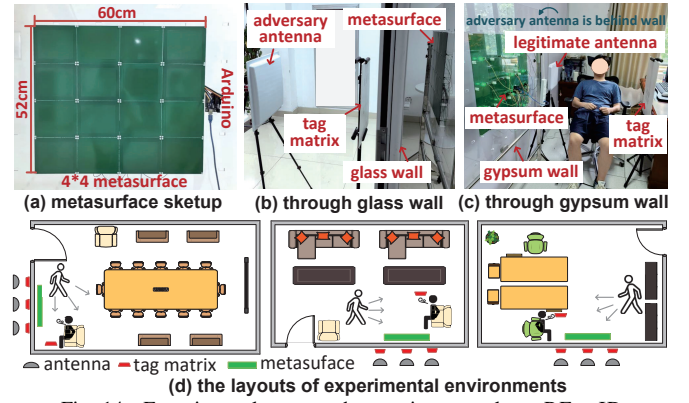


Fig. 14. Experimental setup and scenarios to evaluate RFNOID.

respiration monitoring is marked in Fig. 14(d). Volunteers are asked to walk freely or breathe as usual. RFID tags displayed outside the room are used to detect walking motions, and the tags inside the room are for respiration detection.

3) *Adversary sensing algorithm details*: For motion detection, the variance-based method in [3] is adopted, which sets the threshold as the sum of phase variance *mean* and $7 \times \text{std}$. In the learning-based method, handcrafted features from both temporal and spectral domains listed in [31, 32] are extracted. In addition, we employ the auto-encoder network to extract a latent vector from phase changes. Then, the one-class SVM is used to train the motion detection model. For variance-based and learning-based methods, we collect 30 min signals in advance to extract the threshold and train the detection model. For respiration detection, the FFT method is mostly used to estimate the respiration rate [8, 26].

4) *Baselines*: We compare the RFNOID performance with three baseline metasurface controlling methods: (1) Base strategy, (2) Base strategy + refinement 1, and (3) the strategy in [22], a countermeasure against adversarial WiFi motion detection using a 16×16 WiFi metasurface. Note that [22] only considers the temporal domain signal obfuscation.

5) *Evaluation Metrics*: We adopt two metrics to evaluate walking motion detection, i.e., true positive (TP) and false positive (FP). TP represents the probability of correctly detecting motions when they are present. FP denotes the probability of erroneously detecting the motion when it does not appear. For respiration rate estimation, we employ the mean absolute error (MAE) between the real and measured rate as the metric. The average performance of all tags is recorded as the final result.

B. Evaluation Results

1) *Overall performance*: We show the RFNOID performance in different environments. First, the TP and FP results in walking motion detection with and without our designed metasurface are listed in Table I. The adversary is expected to have a higher TP for accurate motion detection. Indeed, without the metasurface effect, the TP results of both variance-based and learning-based methods are near 1. However, when the metasurface runs, the TP dramatically decreases below 0.1. Particularly, the average TP in different environments and sensing methods is lowered to 0.06, which means that the adversary fails to detect the walking motion. Meanwhile, the

TABLE I
WALKING MOTION DETECTION PERFORMANCE WITH AND WITHOUT RFNOID IN DIFFERENT ENVIRONMENTS

Method	Environment (1)				Environment (2)				Environment (3)			
	w/o metasurface		w. metasurface		w/o metasurface		w. metasurface		w/o metasurface		w. metasurface	
	TP	FP	TP	FP	TP	FP	TP	FP	TP	FP	TP	FP
Variance threshold	1.00	0.00	0.00	0.00	1.00	0.00	0.013	0.00	1.00	0.00	0.00	0.00
Handcrafted features	1.00	0.02	0.028	0.032	0.98	0.056	0.08	0.007	0.89	0.047	0.09	0.02
Auto-encoder	1.00	0.089	0.039	0.008	1.00	0.028	0.003	0.007	1.00	0.015	0.08	0.00

TABLE II
RESPIRATION RATE ESTIMATION WITH AND WITHOUT RFNOID

	Environment (1)		Environment (2)		Environment (3)	
	w/o MS	w. MS	w/o MS	w. MS	w/o MS	w. MS
MAE	0.026	0.087	0.022	0.081	0.019	0.085

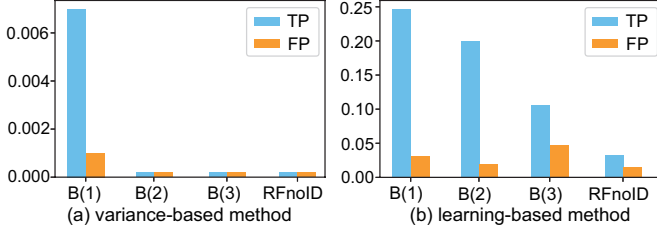


Fig. 15. Comparison with baselines for walking motion detection.

metasurface does not affect the FP results which stay at a low level of around 0. In short, RFNOID can effectively protect the motion presence against the adversary eavesdropping.

The respiration rate estimation results are given in Table II (MS represents metasurface). Without the metasurface, the original MAE of respiration rate in the three environments is 0.022 Hz on average. However, the mean error inclines to 0.084 Hz on average due to the signal obfuscation by the metasurface. In other words, *the respiration rate estimation error is enlarged by around 3 times*, indicating that RFNOID can perturb the respiration signal received by the adversary.

2) *Comparison with baseline metasurface controlling methods*: In this experiment, we compare the performance of RFNOID with three baseline metasurface controlling methods mentioned in Section VI-A4. For walking motion detection, the results using variance-based and learning-based methods are illustrated in Fig. 15. Apart from the B(1) method which only involves the Base strategy, other methods achieve a low TP and FP in the variance-based method, as shown in Fig. 15(a). This is because the Base strategy merely enhances the phase entropy but ignores the requirement of high phase variance. In the learning-based method, the B(2) with refinement 1 and the B(3) for obfuscating temporal WiFi amplitude render higher TPs and FPs than RFNOID, because they overlook the spectral obfuscation of the motion frequency band. Thus, the combination of all parts of the strategy is the most effective, especially overwhelming the existing work [22] that only considers the temporal signal obfuscation.

The MAEs of respiration rate estimation using different methods are depicted in Fig. 16. Similarly, B(1) bears the worst signal obfuscation performance with the least estimation error. B(2) outperforms B(3) due to the enhanced entropy. RFNOID achieves the largest error, verifying the effectiveness of RFNOID in perturbing the regular periodic motion.

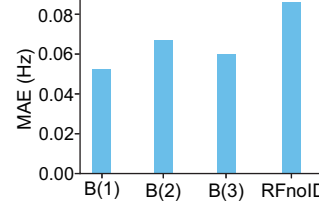


Fig. 16. Comparison with baselines for respiration rate estimation.

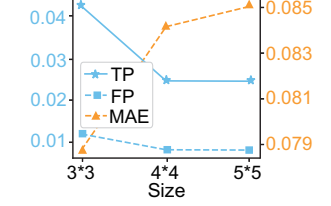


Fig. 17. Impact of metasurface size on RFNOID performance.

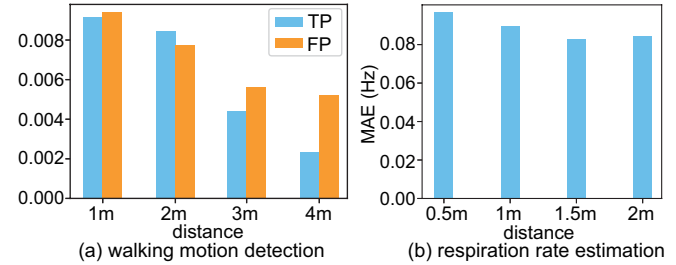


Fig. 18. Impact of the distance between the metasurface and human on RFNOID performance.

3) *Impact of metasurface size*: In this experiment, we investigate the effect of different metasurface sizes, including 3×3 , 4×4 , and 5×5 . We show the walking motion and respiration detection results in Fig. 17. The TP and FP results for walking detection are already below 0.05 when using the small 3×3 metasurface, and the respiration rate error is enlarged to over 0.077 Hz. The 4×4 metasurface can further improve the performance. But the result exhibits only slight enhancement for the 5×5 size. Thus, we use the 4×4 metasurface considering both the fabrication cost and obfuscation performance.

4) *Impact of distance between human and metasurface*: In this experiment, we investigate RFNOID performance under different distances between the human and metasurface. Intuitively, the farther the human body is from the metasurface, the less body-reflected RFID signal may be affected by the metasurface. However, the human body is also farther from the adversary antenna which also receives a weaker signal from the body. Thus, we conduct experiments to explore how the distance affects the motion detection result. As shown in Fig. 18(a), both TP and FP decrease with the increasing distance. This result indicates that even though the metasurface has less influence on the signal reflected by the body at a farther distance, the signal experiences more attenuation to the adversary antenna. In addition, a closer metasurface-human distance also means the body is near the adversary. While RFNOID can still achieve less than 0.009 TP within 1 m.

In terms of respiration detection, the respiration rate MAE in Fig. 18(b) exhibits a slight decline with the increasing metasurface-human distance. This result shows that the far-

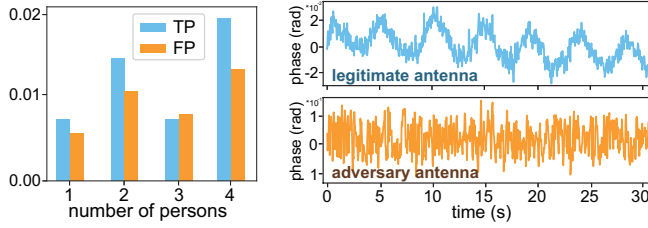


Fig. 19. Impact of the number of moving persons on RFNOID.

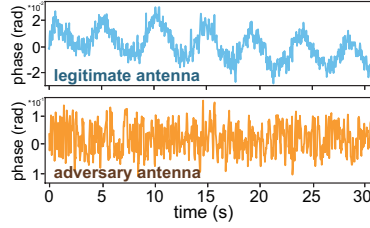


Fig. 20. Respiration phase from legitimate and adversary antennas.

ther metasurface incurs less obfuscation effect on the body-reflected respiration signal. Nevertheless, the respiration estimation error is still above 0.08 Hz when the metasurface-human distance is over 2 m. Thus, the metasurface is effective at different distances to the human body.

5) Impact of the number of persons on motion detection:

In practice, multiple persons may move simultaneously inside the room, causing larger signal phase changes. Accordingly, the metasurface ought to mask the multi-person motion signal. In this experiment, we investigate RFNOID performance with different numbers of moving persons. The TP and FP results for walking motion detection of different numbers of persons are depicted in Fig. 19. The TP and FP illustrate an incline trend with more moving persons because more RFID signals are reflected. While such an incline is not noticeable. The TP and FP are still below 0.02. Thus, RFNOID can effectively obfuscate the motion signals from multiple persons.

6) Impact of metasurface on legitimate RFID sensing:

A key question using RFNOID for privacy-preserving RFID sensing is whether the metasurface will affect the legitimate RFID sensing function inside the room. As shown in Fig. 2(b), the RFID tags inside the room are utilized to monitor the elderly person's respiration. The metasurface should not affect the original RFID sensing task.

Thus, we investigate the impact of metasurface on legitimate RFID sensing. Specifically, we deploy the legitimate RFID antenna beside the person, as depicted in Fig. 14(c). Then, we simultaneously measure the RFID signal from the legitimate and adversary antennas with the metasurface running. As shown in Fig. 20, the phase from the legitimate antenna exhibits a clear and periodic respiration pattern, while the adversary one is obfuscated. Thus, the metasurface does not affect legitimate RFID sensing. This is because the metasurface facing the wall does not reflect the signal transmitted from the legitimate antenna thanks to its directivity.

VII. RELATED WORKS

This section introduces the related works in studying privacy issues of RFID and RF human sensing.

A. Privacy-concerned RFID Human Motion Sensing

RFID-based human motion sensing has raised significant privacy concerns, especially due to its through-wall sensing capability [13]. RFID signals mainly expose three types of private information, i.e., human presence and location, living habits, and biometrics [33]. For example, Tadar [4] and RF-HMS [3] manage to detect and track the walking motion through the wall. TASA [34] and FEMO [12] realize contactless human daily activity monitoring by deploying

RFID tags in the environment. Many physiological features, e.g., respiration [9] and facial dynamics [35], and behavioral features, e.g., gait [36] and handshake [37], can be captured by RFID signals for human authentication and identification. In addition, researchers have demonstrated the feasibility of using commodity RFID devices for through-wall eavesdropping [15, 17]. Thus, it is crucial to implement privacy measures to secure RFID sensing systems. However, existing studies mainly focus on securing RFID communication [20, 38] but overlook the privacy issues in RFID-based motion sensing. In this work, we fill this gap by proposing RFNOID to protect human motion privacy efficiently.

B. Privacy-preserving RF Human Sensing

Apart from RFID signals, other wireless signals, like WiFi and FMCW radar, also face privacy leakage issues. Researchers are intensively investigating potential attacks on existing RF sensing systems. On the one hand, WiFi and radar signals are exploited as adversarial sensors to directly detect through-wall human presence [39, 40], habitual behaviors [41], and posture tracking [42]. On the other hand, physical adversarial attacks are conducted to destroy the original sensing functions by jamming the RF signal [43, 44]. Thereby, countermeasures are developed to safeguard RF-based human sensing systems. Secur-Fi [45] actively controls the WiFi transmitting antenna to jam the signal perceived by the adversary. IRShield [22] and RISTealth [23] employ the metasurface to passively overwhelm or hide the human motion WiFi signal in the temporal domain. RF-Protect [21] designs a radar signal reflector to generate fake motion patterns.

However, existing WiFi and radar solutions [21–23] that are designed for obfuscating WiFi signal amplitude and radar frequency cannot be directly borrowed by RFID sensing systems in which the sensitive signal phase is mostly employed for motion sensing. Besides, the metasurface strategies in [22, 23] merely focus on masking the temporal human signal pattern. Instead, RFNOID develops a dedicated signal phase obfuscation strategy for both temporal and spectral domains.

VIII. CONCLUSION

In this paper, we introduce RFNOID, the first system designed to protect human motion privacy from adversarial through-wall RFID sensing. Central to our solution is using the metasurface to obfuscate the channel to protect the motion information. We envision that RFNOID will raise the bar for privacy protection against through-wall sensing systems and anticipate its integration into future smart surfaces to safeguard privacy across various RF sensing modalities.

IX. ACKNOWLEDGEMENT

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